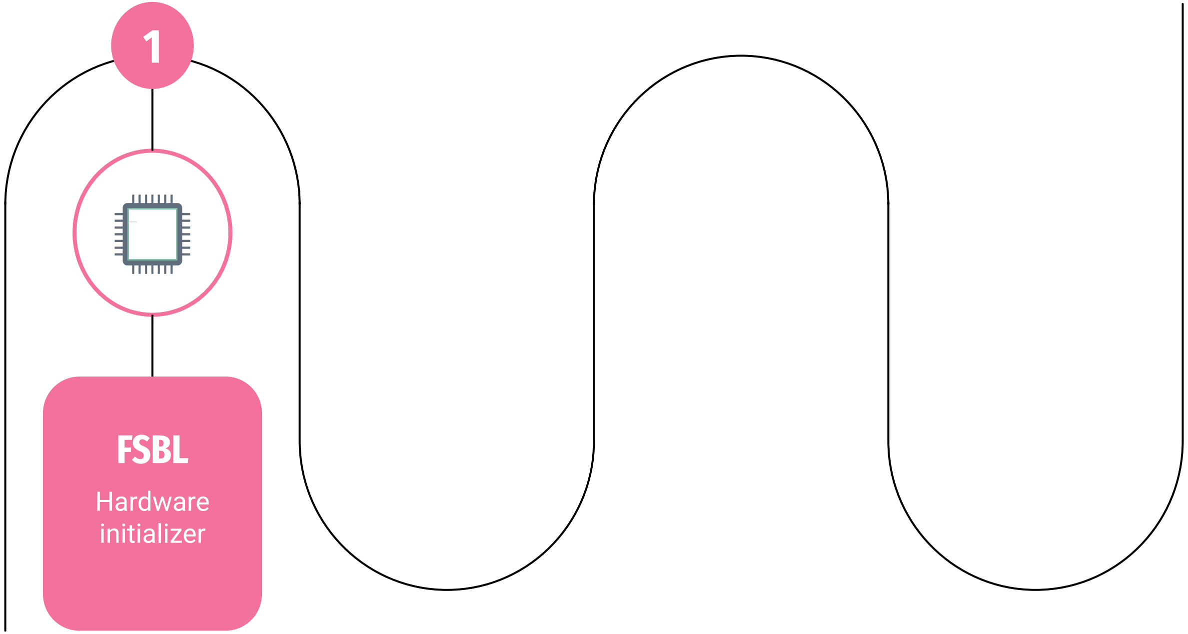
Linux boot on SHAKTI

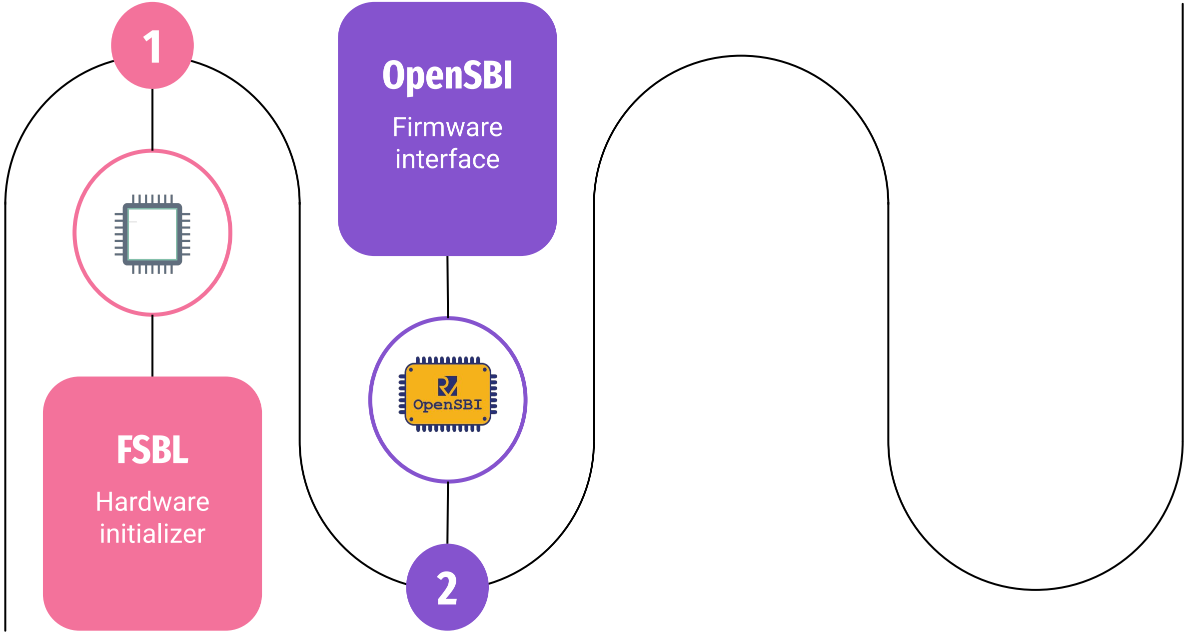
SHAKTI Group | CSE Dept | PS-CDISHA - RISE Lab | IIT Madras

Linux boot on SHAKTI

Linux boot on SHAKTI



Linux boot on SHAKTI



Linux boot on SHAKTI

1. **FSBL**
   Hardware initializer

2. **OpenSBI**
   Firmware interface

3. **U-boot**
   Universal bootloader - bridge b/w firmware and OS

Linux boot on SHAKTI

1

FSBL

Hardware initializer

2

OpenSBI

Firmware interface

3

U-boot

Universal bootloader - bridge b/w firmware and OS

4

Kernel & Buildroot

OS managing Hardware resources and filesystem

Linux boot on SHAKTI

1

FSBL

Hardware initializer

2

OpenSBI

Firmware interface

3

U-Boot

U-boot

Universal bootloader - bridge b/w firmware and OS

4

Kernel & Buildroot

OS managing hardware resources and filesystem

Build Root
Making Embedded Linux Easy

SHAKTI

Steps to Boot

Prerequisites

- Ubuntu/Linux host machine
- Installed dependencies: git, make, gcc, riscv64-unknown-elf-gcc, openocd, gdb
- SD card and Arty 100T FPGA board with SPI SD card module

Clone the Repository and Do the Intial steps

```
$ git clone  https://gitlab.com/shaktiproject/software/linux-devkit

$ cd linux-devkit

$ 19-sd-support-with-python-and-pip

$ git submodule update --init --recursive
```

Steps to Boot

Run the following too generate the Linux Image

```
$ make image -j $(nprocs)
```

The build process automatically:

- Builds OpenSBI
- Builds U-Boot
- Builds Buildroot (root filesystem)
- Builds Linux kernel

The process can take several minutes.

Output Files

After a successful build, binaries are placed in the output/ directory:

```
output/
├── code.mem
├── uImage
├── rootfs.tar
├── shakti.dtb
└── fw_payload.elf
```

Prepare the SD Card

Partition the SD card into three partitions:

Partition	Label	Content
1	RFS	Root filesystem files
2	BOOT	uImage and system.dtb (inside boot folder)
3	CODE	code.mem

Steps:

- Use fdisk or gparted to create 3 partitions.
- Format them appropriately (e.g., ext4 for RFS,BOOT fat32 for CODE).

Prepare the SD Card (Contd)

Mount and copy files:

```
$ sudo cp <linux-devkit_dir_path>/output/rootfs.tar /media/<user>/RFS/

$ sudo mkdir /media/<user>/BOOT/boot

$ sudo cp <linux-devkit_dir_path>/output/uImage /media/<user>/BOOT/boot

$ sudo cp <linux-devkit_dir_path>/output/shakti.dtb /media/<user>/BOOT/boot

$ sudo cp code.mem /media/<user>/CODE/
```

Run following commands for permission access:

```
For both BOOT and RFS

$ sudo chmod -R 777 *

$ sudo chown -R root *
```

Booting Linux

- Insert the prepared SD card into the SD module connected via SPI.
- Power up the SHAKTI board.
- Connect via OpenOCD and GDB:

```
$ sudo $(which openocd) -f bsp/third_party/ganga/ftdi.cfg
In another terminal
$ riscv64-unknown-elf-gdb
(gdb) target remote : 3333
(gdb) file <path_to_output_dir>/fw_payload.elf
(gdb) load
(gdb) c
In another terminal to see output log of booted linux
$ screen /dev/ttyUSBx 19200
```


SHAKTI

Output

- Here the OpenSBI will prompt with press any key, press enter and type boot and enter

```
OpenSBI v0.9-23-ge71a7c1
   ____                    _____ ____ _____
  / __ \                  / ____|  _ \_   _|
 | |  | |_ __   ___ _ __ | (___ | |_) || |
 | |  | | '_ \ / _ \ '_ \ \___ \|  _ < | |
 | |__| | |_) |  __/ | | |____) | |_) || |_
  \____/| .__/ \___|_| |_|_____/|____/_____|
        | |
        |_|

Platform Name             : ucbbar,spike-bare
Platform Features         : timer,mfdeleg
Platform HART Count       : 1
Firmware Base             : 0x80000000
Firmware Size             : 100 KB
Runtime SBI Version       : 0.3

Domain0 Name              : root
Domain0 Boot HART         : 0
Domain0 HARTs             : 0*
Domain0 Region00          : 0x0000000080000000-0x000000008001ffff ()
Domain0 Region01          : 0x0000000000000000-0xffffffffffffffff (R,W,X)
Domain0 Next Address      : 0x0000000080200000
Domain0 Next Arg1         : 0x0000000082200000
Domain0 Next Mode         : S-mode
Domain0 SysReset          : yes

Boot HART ID              : 0
Boot HART Domain          : root
Boot HART ISA             : rv64imafdcsu
Boot HART Features        : scounteren,mcounteren
Boot HART PMP Count       : 16
Boot HART PMP Granularity : 4
Boot HART PMP Address Bits: 54
Boot HART MHPM Count      : 0
Boot HART MHPM Count      : 0
Boot HART MIDELEG         : 0x0000000000000222
Boot HART MEDELEG         : 0x000000000000b109
```

Output

```
[    2.876464] futex hash table entries: 256 (order: 0, 6144 bytes, linear)
[    3.729583] clocksource: Switched to clocksource riscv_clocksource
[   28.557861] workingset: timestamp_bits=62 max_order=16 bucket_order=0
[   45.327362] random: get_random_bytes called from init_oops_id+0x38/0x4c with crng_init=0
[   48.244750] Freeing unused kernel memory: 9440K
[   48.383514] This architecture does not have kernel memory protection.
[   48.636383] Run /init as init process
Starting syslogd: OK
Starting klogd: OK
Running sysctl: OK
Starting mdev... OK
modprobe: can't change directory to '/lib/modules': No such file or directory
Starting network: ip: socket: Function not implemented
ip: socket: Function not implemented
FAIL
Starting dropbear sshd: FAIL

Welcome to Buildroot
buildroot login: root

Password:
login[64]: root login on 'console'
#
# ls
#
# pwd
/root
#
# cd /
#
# ls
bin       init      linuxrc   opt       run       tmp
dev       lib       media     proc      sbin      usr
etc       lib64     mnt       root      sys       var
#
# 
```

Thank you

Website: shakti.org.in

GitLab: https://gitlab.com/shaktiproject/software/linux-devkit